Chapter 2

Placing Data Science within the Realm of AI

IN THIS CHAPTER

- **Understanding how data and data science relate**
- **Considering the progression into AI and beyond**
- **Developing a data pipeline to AI**

Some people perceive data science as simply a method of managing data for use with an AI discipline, but you can use your data science skills for a great many tasks other than AI. You use data science skills for various types of statistical analysis that don't rely on an AI, such as to perform analytics, manage data in various ways, and locate information that you use directly rather than as an input into anything. However, the data science to AI connection does exist as well, so you need to know about it as a data scientist, which is the focus of the first part of this chapter.

Many terms used in data science become muddled because people misuse them. When you hear the term *AI,* you might think about all sorts of technologies that are either distinct AI subcategories or have nothing to do with AI at all. The second part of the chapter defines AI and then clarifies its connection to machine learning, which is a subcategory of AI, and finally to deep learning, which is actually a subcategory of machine learning. Understanding this hierarchy is important in understanding the role data science plays in making these technologies work.

The first two sections define the endpoints of a data pipeline. The third section describes the pipeline between data science and AI (and its subcategories). This data pipeline is a particular implementation of data science skills, so you need to know about it. You also need to consider that this data pipeline isn't the only one you need to create and use as a data scientist. For example, you might be involved in a type of data mining that doesn't rely on AI but rather on specific sorts of filtering, sorting, and the use of statistical analysis. The articles at `https://www.innoarchitech.com/blog/data-science-big-data-explained-non-data-scientist` and `https://www.northeastern.edu/graduate/blog/what-does-a-data-scientist-do/` give you some other ideas about how data scientists create and use data pipelines.

Seeing the Data to Data Science Relationship

Obviously, to become a data scientist, you must have data to work with. What isn't obvious is the kind of data, the data sources, and the uses of the data. Data is the requirement for analysis, but that analysis can take many forms. For example, the article at `https://blog.allpsych.com/spending-habits-can-reveal-personality-traits/` talks about data used to guess your psychological profile, which can then be used for all sorts of purposes — many positive; others not. The issue is that these analyses often help others know more about you than you know yourself, which is a scary thought when you consider how someone might use the information.

The following sections discuss data, not the moral or ethical complications of gathering it. In general, you find that some analyses work fine with just a little data, but others require huge amounts of it. For example, later sections discuss AI, machine learning, and deep learning, which require ascending amounts of data to perform well.

Considering the data architecture

A data architecture consists of a number of elements. Although you may not always have to deal with these elements as a data scientist, you should know about them to create robust solutions:

- Format
- Speed of access
- Cost of access
- Industry
- Use

Data comes in many forms. Book 2 discusses all sorts of architectures in detail — everything from in-memory databases to those found in huge corporate databases. The form of the data differs as well. A Relational Database Management System (RDBMS) provides a highly structured dataset that's easy to interact with, but it places restrictions on the data form. On the other hand, a Not Only SQL (NoSQL) database is freeform, so it's flexible in a way that an RDBMS isn't. The form of the data determines what you need to do to manicure it for your particular need, because raw data rarely comes in precisely the format you require.

The data architecture also considers how you access the data. Even though online data sources are convenient and accessible from just about everywhere, they come with a speed penalty that you must consider when using them. Localized databases are faster, but now you have to deal with data that might be stale or not as pertinent to a particular need. You also risk losing the true world view as you focus on data that comes from your organization.

Some people are under the assumption that the quality of data is the same no matter what the source. Quality does vary, however, and you need to consider how the quality of your data affects your analysis. When data appears in the public domain, it may not be very high quality because no one has vetted it. This isn't to say that you can't find high-quality, public-domain data — just that the likelihood is less. Likewise, data you pay to access isn't always guaranteed to meet your specific needs even if it is high quality. The cost of access affects your analysis because the tendency is to reduce costs by taking the penalty in quality.

The industry generating or using the data also matters. You can become blindsided by the biases of your particular industry. The data you need to perform a financial analysis may actually reside in the data generated by the energy industry. Yet, if you don't look in that particular industry's offerings, you won't ever find this hidden data, and your analysis will fail or at least produce less robust results.

WARNING Perhaps the most problematic component of data architecture, however, is how you use the data you obtain. Logic errors occur all the time, and having a list of best practices might seem as though it could provide a solution to the problem, but it can't. Intuition and strong logical analysis often form a part of the most successful data usage. This means that if you perform a survey and obtain results that seem ludicrous, they probably are, and you need to review the data again, rather than publish the sensational results you think you might have obtained from a flawed analysis.

Acquiring data from various sources

A data source need not come in the form of a huge corporate database. Some of the more interesting forms of analysis use mundane and tiny datasets derived from sources such as Point-of-Sale (POS) systems. The article at `https://blog.magestore.com/pos-data/` makes a strong case for analyzing POS data for more than simply knowing when to re-order certain items and determining sales on a given day. Some datasets are public domain and free (often used for testing and training a deep learning application) and others require a purchase. Some sites actually sell data as a commodity, such as Data World (`https://data.world/datasets/wildlife`). The point is that you're drowning in a sea of data and may not even know it.

The source of the data also determines a great many details about what you can expect. For example, POS data can prove hard to analyze because the terminals lack standardization for the most part, and you can't be sure precisely what form the data will take. If you have terminals from various vendors, each of whom uses a different data format, the problem is compounded because you now need custom cleaning solutions for each of the data sources.

REMEMBER A data source may be unreliable because it is accessible only during certain hours of the day or might experience unexpected down times. For example, consider getting data from a rover on Mars. All sorts of issues can occur, as described in the article at `https://www.theatlantic.com/science/archive/2018/09/curiosity-opportunity-nasa-rover-problems/570769/`. Of course, most people aren't working with a rover on Mars, but corporate databases can go offline as well. Lest you think that the cloud fixes reliability, check out the article at `https://www.networkworld.com/article/3394341/when-it-comes-to-uptime-not-all-cloud-providers-are-created-equal.html`. In short, you always need a Plan B when it comes to data, and having a Plan C is likely a good idea as well.

Performing data analysis

At some point, every data scientist performs an analysis, which means working with the data to produce a usable result. The analysis could be anything. You might simply want to know the average of sales for March over the last ten years. Whatever the reason for the analysis, the results you achieve depend on these issues:

- The reliability of the data
- The quality of the data
- The quantity of the data
- Selection of the right algorithm
- Presentation of the result in the correct manner
- Interpretation of the result
- Tuning of the result

TIP As you proceed through the book, starting with Book 2, you find that all these issues come under discussion for various data sources, algorithms, and analysis techniques. However, always keep in mind that automation never trumps human insight and intuition. As you go through the book, remember that you must ultimately decide whether a result is correct (at least, correct enough) and useful.

Archiving the data

Archiving data is important for historical reasons. You want to have the data available to repeat a particular analysis. A new algorithm might reveal something different about the data, or you may have to subject your analysis method to peer review. Only when the data becomes outdated do you consider getting rid of it. Unlike a shirt that you accidentally send to a resale store but may still be able to buy back, you can't retrieve your data after it's gone.

Most people will find it easy to argue that static data, the type you collect once or at specific intervals, requires archiving. Part of the reason is that static data is controllable. You won't suddenly find all your storage devices cluttered with it.

REMEMBER Some sections of this book deal with live data — the sort that changes minute by minute. Archiving such data might seem to be useless. You might argue that no one will actually need the data you collected five minutes ago. However, just as police use recordings of live camera feeds to find the culprit in a burglary or a carjacking, you need to archive live data to locate the source of issues in your organization. Always archive all your data even if the archive lasts only a short time.

Defining the Levels of AI

AI is a broad term for a set of technologies that use data in a particular way to perform specific kinds of analysis. The underlying assumption is that a computer will automate the process of performing this analysis through algorithms that manipulate the data in distinctive ways. When you look at AI, you see a high-level technology that leads to refined technologies such as machine learning. Machine learning isn't AI; instead, it's a subset of AI, and keeping the distinction clear is important. Likewise, one refinement of machine learning is deep learning. Again, deep learning isn't machine learning, and you need to keep them separate. The following sections offer definitions for these three terms and help you understand them.

Beginning with AI

Before you can use a term in any meaningful and useful way, you must have a definition for it. After all, if nobody agrees on a meaning, the term has none; it's just a collection of characters. Defining the idiom (which is a term whose meaning isn't clear from the meanings of its constituent elements) is especially important with technical terms that have received more than a little press coverage at various times and in various ways.

REMEMBER Saying that AI is an artificial intelligence doesn’t really tell you anything meaningful, which is why so many discussions and disagreements arise over this term. Yes, you can argue that what occurs is artificial, not having come from a natural source. However, the intelligence part is, at best, ambiguous. Even if you don’t necessarily agree with the definition of AI as it appears in the sections that follow, this book uses AI according to that definition, and knowing it will help you follow the rest of the text more easily.

Discerning intelligence

People define intelligence in many different ways. However, you can say that intelligence involves certain mental activities composed of the following actions:

- **Learning:** Having the ability to obtain and process new information
- **Reasoning:** Being able to manipulate information in various ways
- **Understanding:** Considering the result of information manipulation
- **Grasping truths:** Determining the validity of the manipulated information
- **Seeing relationships:** Divining how validated data interacts with other data
- **Considering meanings:** Applying truths to particular situations in a manner consistent with their relationship
- **Separating fact from belief:** Determining whether the data is adequately supported by provable sources that can be demonstrated to be consistently valid

The list could easily get quite long, but even this list is relatively prone to interpretation by anyone who accepts it as viable. As you can see from the list, however, intelligence often follows a process that a computer system can mimic as part of a simulation:

1. Set a goal based on needs or wants.
2. Assess the value of any currently known information in support of the goal.
3. Gather additional information that could support the goal.

4. Manipulate the data such that it achieves a form consistent with existing information.
5. Define the relationships and truth values between existing and new information.
6. Determine whether the goal is achieved.
7. Modify the goal in light of the new data and its effect on the probability of success.
8. Repeat Steps 2 through 7 as needed until the goal is achieved (found true) or the possibilities for achieving it are exhausted (found false).

REMEMBER Even though you can create algorithms and provide access to data in support of this process within a computer, a computer's capability to achieve intelligence is severely limited. For example, a computer is incapable of understanding anything because it relies on machine processes to manipulate data using pure math in a strictly mechanical fashion. Likewise, computers can't easily separate truth from mistruth (as described in Book 6, Chapter 2). In fact, no computer can fully implement any of the mental activities described in the list that describes intelligence.

As part of deciding what intelligence actually involves, categorizing intelligence is also helpful. Humans don't use just one type of intelligence, but rather rely on multiple intelligences to perform tasks. Howard Gardner of Harvard has defined a number of these types of intelligence (see `http://www.pz.harvard.edu/projects/multiple-intelligences` for details), and knowing them helps you to relate them to the kinds of tasks that a computer can simulate as intelligence. Here is a modified version of these intelligences with additional description:

- **Visual-spatial:** Physical environment intelligence used by people like sailors and architects (among many others). To move at all, humans need to understand their physical environment — that is, its dimensions and characteristics. Every robot or portable computer intelligence requires this capability, but the capability is often difficult to simulate (as with self-driving cars) or less than accurate (as with vacuums that rely as much on bumping as they do on moving intelligently).

- **Bodily-kinesthetic:** Body movements, such as those used by a surgeon or a dancer, require precision and body awareness. Robots commonly use this kind of intelligence to perform repetitive tasks, often with higher precision than humans, but sometimes with less grace. It's essential to differentiate between human augmentation, such as a surgical device that provides a surgeon with enhanced physical ability, and true independent movement. The former is simply a demonstration of mathematical ability in that it depends on the surgeon for input.
- **Creative:** Creativity is the act of developing a new pattern of thought that results in unique output in the form of art, music, and writing. A truly new kind of product is the result of creativity. An AI can simulate existing patterns of thought and even combine them to create what appears to be a unique presentation but is really just a mathematically based version of an existing pattern. In order to create, an AI would need to possess self-awareness, which would require intrapersonal intelligence.
- **Interpersonal:** Interacting with others occurs at several levels. The goal of this form of intelligence is to obtain, exchange, give, and manipulate information based on the experiences of others. Computers can answer basic questions because of keyword input, not understanding. The intelligence occurs while obtaining information, locating suitable keywords, and then giving information based on those keywords. Cross-referencing terms in a lookup table and then acting upon the instructions provided by the table demonstrates logical intelligence, not interpersonal intelligence.
- **Intrapersonal:** Looking inward to understand one's own interests and then setting goals based on those interests is currently a human-only kind of intelligence. As machines, computers have no desires, interests, wants, or creative abilities. An AI processes numeric input using a set of algorithms and provides an output — it isn't aware of anything that it does, nor does it understand anything that it does.
- **Linguistic:** Working with words is an essential tool for communication because spoken and written information exchange is far faster than any other form. This form of intelligence includes understanding spoken and written input, managing the input to develop an answer, and providing an understandable answer as output. In many cases, computers can barely parse input into keywords, can't actually understand the request at all, and output responses that may not be understandable at all. In humans, spoken and written linguistic intelligence come from different areas of the brain

(`https://releases.jhu.edu/2015/05/05/say-what-how-the-brain-separates-our-ability-to-talk-and-write/`), which means that even with humans, someone who has high written linguistic intelligence may not have similarly high spoken linguistic intelligence. Computers don't currently separate written and spoken linguistic ability.

- **Logical-mathematical:** Calculating a result, performing comparisons, exploring patterns, and considering relationships are all areas in which computers currently excel. When you see a computer beat a human on a game show, this is the only form of intelligence, out of seven, that you're actually seeing. Yes, you might see small bits of other kinds of intelligence, but this one is the focus. Basing an assessment of human versus computer intelligence on just one area isn't a good idea.

Discovering four ways to define AI

As described in the previous section, the first concept that's important to understand is that AI doesn't really have anything to do with human intelligence. Yes, some AI is modeled to simulate human intelligence, but that's what it is: a simulation. When thinking about AI, notice an interplay between goal seeking, data processing used to achieve that goal, and data acquisition used to better understand the goal. AI relies on algorithms to achieve a result that may or may not have anything to do with human goals or methods of achieving those goals. With this understanding in mind, you can categorize AI in four ways:

- **Acting humanly:** When a computer acts like a human, it best reflects the Turing test, in which the computer succeeds when differentiation between the computer and a human isn't possible (see `https://www.turing.org.uk/scrapbook/test.html` for details). This category also reflects what the media would have you believe AI is all about. You see it employed for technologies such as natural language processing, knowledge representation, automated reasoning, and machine learning (all four of which must be present to pass the test).
- **Thinking humanly:** When a computer thinks as a human, it performs tasks that require intelligence (as contrasted with rote procedures) from a human to succeed, such as driving a car. To determine whether a program thinks like a human, you must have some method of determining how humans

think, which the cognitive modeling approach defines. This model relies on three techniques that are used to create a model, which forms the basis for a simulation:

- **Introspection:** Detecting and documenting the techniques used to achieve goals by monitoring one's own thought processes.
- **Psychological testing:** Observing a person's behavior and adding it to a database of similar behaviors from other persons given a similar set of circumstances, goals, resources, and environmental conditions (among other things).
- **Brain imaging:** Monitoring brain activity directly through various mechanical means, such as Computerized Axial Tomography (CAT), Positron Emission Tomography (PET), Magnetic Resonance Imaging (MRI), and Magnetoencephalography (MEG).

- **Thinking rationally:** Applying some type of standard to studying how humans think enables the creation of guidelines that describe typical human behaviors. A person is considered rational when following these behaviors within certain levels of deviation. A computer that thinks rationally relies on the recorded behaviors to create a guide as to how to interact with an environment based on the data at hand. The goal of this approach is to solve problems logically, when possible. In many cases, this approach would enable the creation of a baseline technique for solving a problem, which would then be modified to actually solve the problem. In other words, the solving of a problem in principle is often different from solving it in practice, but you still need a starting point.
- **Acting rationally:** Studying how humans act in given situations under specific constraints enables you to determine which techniques are both efficient and effective. A computer that acts rationally relies on the recorded actions to interact with an environment based on conditions, environmental factors, and existing data. As with rational thought, rational acts depend on a solution in principle, which may not prove useful in practice. However, rational acts do provide a baseline upon which a computer can begin negotiating the successful completion of a goal.

Using categories to define kinds of AI

The categories used to define AI offer a way to consider various uses for or ways to apply AI. Some of the systems used to classify AI by type are arbitrary and not distinct. For example, some groups view AI as either strong (generalized intelligence that can adapt to a variety of situations) or weak (specific intelligence designed to perform a particular task well). The problem with strong AI is that it doesn't perform any task well, while weak AI is too specific to perform tasks independently. Even so, having just two type classifications won't do the job even in a general sense. The four classification types promoted by Arend Hintze (see `http://theconversation.com/understanding-the-four-types-of-ai-from-reactive-robots-to-self-aware-beings-67616` for details) form a better basis for understanding AI:

- **Reactive machines:** The machines you see beating humans at chess or playing on game shows are examples of reactive machines. A reactive machine has no memory or experience upon which to base a decision. Instead, it relies on pure computational power and smart algorithms to re-create every decision every time. This is an example of a weak AI used for a specific purpose.
- **Limited memory:** A self-driving car or autonomous robot can't afford the time to make every decision from scratch. These machines rely on a small amount of memory to provide experiential knowledge of various situations. When the machine sees the same situation, it can rely on experience to reduce reaction time and to provide more resources for making new decisions that haven't yet been made. This is an example of the current level of strong AI.
- **Theory of mind:** A machine that can assess both its required goals and the potential goals of other entities in the same environment has a kind of understanding that is feasible to some extent today, but not in any commercial form. However, for self-driving cars to become truly autonomous, this level of AI must be fully developed. A self-driving car would not only need to know that it must go from one point to another, but also intuit the potentially conflicting goals of drivers around it and react accordingly.
- **Self-awareness:** This is the sort of AI that you see in movies. However, it requires technologies that aren't even remotely possible now because such a machine would have a sense of both self and consciousness. In addition, instead of merely intuiting the goals of others based on environment and other entity reactions, this type of machine would be able to infer the intent of others based on experiential knowledge.

Advancing to machine learning

Machine learning is a specific application of AI used to simulate human learning through algorithms developed using various techniques, such as matching a series of inputs to a desired set of outputs. The following sections explore machine learning as it applies to this book.

Exploring what machine learning can do for AI

Machine learning relies on algorithms to analyze huge datasets. Currently, machine learning can't provide the sort of AI that the movies present. Even the best algorithms can't think, feel, present any form of self-awareness, or exercise free will. What machine learning can do is perform predictive analytics far faster than any human can. As a result, machine learning can help humans work more efficiently. The current state of AI, then, is one of performing analysis, but humans must still consider the implications of that analysis — making the required moral and ethical decisions. The essence of the matter is that machine learning provides just the learning part of AI, and that part is nowhere near ready to create an AI of the sort you see in films.

REMEMBER The main point of confusion between learning and intelligence is that people assume that simply because a machine gets better at its job (learning), it's also aware (intelligence). Nothing supports this view of machine learning. The same phenomenon occurs when people assume that a computer is purposely causing problems for them. The computer can't assign emotions and therefore acts only upon the input provided and the instruction contained within an application to process that input. A true AI will eventually occur when computers can finally emulate the clever combination used by nature:

- **Genetics:** Slow learning from one generation to the next
- **Teaching:** Fast learning from organized sources
- **Exploration:** Spontaneous learning through media and interactions with others

Considering the goals of machine learning

At present, AI is based on machine learning, and machine learning is essentially different from statistics. Yes, machine learning has a statistical basis, but it makes some different assumptions than statistics do because the goals are different. Table 2-1 lists some features to consider when comparing AI and machine learning to statistics.

TABLE 2-1: Comparing Machine Learning to Statistics

Technique	***Machine Learning***	***Statistics***
Data handling	Works with big data in the form of networks and graphs; raw data from sensors or web text is split into training and test data.	Models are used to create predictive power on small samples.
Data input	The data is sampled, randomized, and transformed to maximize accuracy scoring in the prediction of out-of-sample (or completely new) examples.	Parameters interpret real-world phenomena and focus on magnitude.
Result	Probability is taken into account for comparing what could be the best guess or decision.	The output captures the variability and uncertainty of parameters.

Technique	***Machine Learning***	***Statistics***
Assumptions	The scientist learns from the data.	The scientist assumes a certain output and tries to prove it.
Distribution	The distribution is unknown or ignored before learning from data.	The scientist assumes a well-defined distribution.
Fitting	The scientist creates a best-fit, but generalizable, model.	The result is fit to the present data distribution.

Defining machine learning limits based on hardware

Huge datasets require huge amounts of memory. Unfortunately, the requirements don't end there. When you have huge amounts of data and memory, you must also have processors with multiple cores and high speeds. One of the problems that scientists are striving to solve is how to use existing hardware more efficiently. In some cases, waiting for days to obtain a result to a machine learning problem simply isn't possible. The scientists who want to know the answer need it quickly, even if the result isn't quite right. In addition, investments in better hardware also require investments in better science. This book considers some of the following issues as part of making your machine learning experience better:

- **Obtaining a useful result:** As you work through the book, you discover that you need to obtain a useful result first, before you can refine it. In addition, sometimes tuning an algorithm goes too far and the result becomes quite fragile (and possibly useless outside a specific dataset).
- **Asking the right question:** Many people get frustrated in trying to obtain an answer from machine learning because they keep tuning their algorithm

without asking a different question. To use hardware efficiently, sometimes you must step back and review the question you're asking. The question might be wrong, which means that even the best hardware will never find the answer.

- **Relying on intuition too heavily:** All machine learning questions begin as a hypothesis. A scientist uses intuition to create a starting point for discovering the answer to a question. Failure is more common than success when working through a machine learning experience. Your intuition adds the art to the machine learning experience, but sometimes intuition is wrong, and you have to revisit your assumptions.

Considering the true uses of AI and machine learning

You find AI and machine learning used in a great many applications today. The only problem is that the technology works so well that you don't know that it even exists. In fact, you might be surprised to find that many devices in your home already make use of both technologies. Both technologies definitely appear in your car and most especially in the workplace. In fact, the uses for both AI and machine learning number in the millions — all safely out of sight even when they're quite dramatic in nature. Here are just a few of the ways in which you might see AI and machine learning used together:

- **Fraud detection:** You get a call from your credit card company asking whether you made a particular purchase. The credit card company isn't being nosy; it's simply alerting you to the fact that someone else could be making a purchase using your card. The AI embedded within the credit card company's code detected an unfamiliar spending pattern and alerted someone to it.
- **Resource scheduling:** Many organizations need to schedule the use of resources efficiently. For example, a hospital may have to determine where to put a patient based on the patient's needs, availability of skilled experts, and the amount of time the doctor expects the patient to be in the hospital.
- **Complex analysis:** Humans often need help with complex analysis because there are literally too many factors to consider. For example, the same set of symptoms could indicate more than one problem. A doctor or other expert might need help making a diagnosis in a timely manner to save a patient's life.

- **Automation:** Any form of automation can benefit from the addition of AI to handle unexpected changes or events. A problem with some types of automation today is that an unexpected event, such as an object in the wrong place, can actually cause the automation to stop. Adding AI to the automation can allow the automation to handle unexpected events and continue as if nothing happened.
- **Customer service:** The customer service line you call today may not even have a human behind it. The automation is good enough to follow scripts and use various resources to handle the vast majority of your questions. With good voice inflection (provided by AI as well), you may not even be able to tell that you're talking with a computer.
- **Safety systems:** Many of the safety systems found in machines of various sorts today rely on AI to take over the vehicle in a time of crisis. For example, many automatic braking systems rely on AI to stop the car based on all the inputs that a vehicle can provide, such as the direction of a skid.
- **Machine efficiency:** AI can help control a machine in such a manner as to obtain maximum efficiency. The AI controls the use of resources so that the system doesn't overshoot speed or other goals. Every ounce of power is used precisely as needed to provide the desired services.

This list doesn't even begin to scratch the surface. You can find AI and machine learning combined in many other ways. However, it's also useful to view uses of machine learning outside the normal realm that many consider the domain of AI. Here are a few uses for machine learning that you might not associate with AI:

- **Access control:** In many cases, access control is a yes-or-no proposition. An employee smartcard grants access to a resource in much the same way that people have used keys for centuries. Some locks do offer the capability to set times and dates that access is allowed, but the coarse-grained control doesn't really answer every need. By using machine learning, you can determine whether an employee should gain access to a resource based on role and need. For example, an employee can gain access to a training room when the training reflects an employee role.
- **Animal protection:** The ocean might seem large enough to allow animals and ships to cohabitate without problems. Unfortunately, many animals get hit by ships each year. A machine learning algorithm could allow ships to avoid animals by learning the sounds and characteristics of both the animal and the ship.

- **Predicting wait times:** Most people don't like waiting when they have no idea of how long the wait will be. Machine learning allows an application to determine waiting times based on staffing levels, staffing load, complexity of the problems the staff is trying to solve, availability of resources, and so on.

Getting detailed with deep learning

An understanding of deep learning begins with a precise definition of terms. Otherwise, you have a hard time separating the media hype from the realities of what deep learning can actually provide. Deep learning is part of both AI and machine learning, as shown in Figure 2-1. To understand deep learning, you must begin at the outside — that is, you start with AI, and then work your way through machine learning, and then finally define deep learning. The following sections help you through this process.

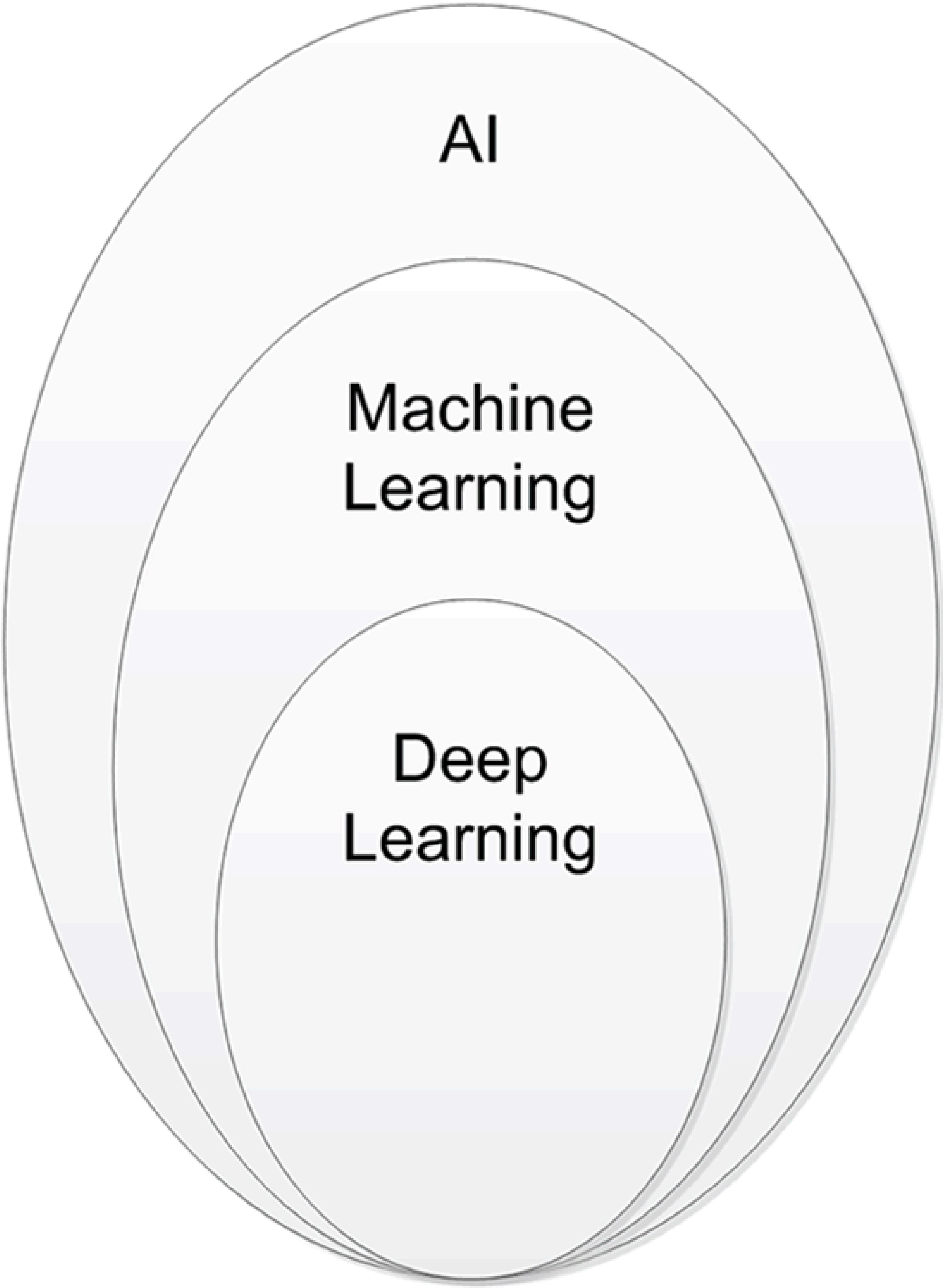

FIGURE 2-1: Deep learning is a subset of machine learning which is a subset of AI.

Moving from machine learning to deep learning

Deep learning is a subset of machine learning, as previously mentioned. In both cases, algorithms appear to learn by analyzing huge amounts of data, although learning can occur even with tiny datasets in some cases. However, deep learning varies in the depth of its analysis and the kind of automation it provides. You can summarize the differences between the two like this:

- **A completely different paradigm:** Machine learning consists of a set of many different techniques that enable a computer to learn from data and to use what it learns to provide an answer, often in the form of a prediction. Machine learning relies on different paradigms, such as using statistical analysis, finding analogies in data, using logic, and working with symbols. Contrast the myriad techniques used by machine learning with the single technique used by deep learning, which mimics human brain functionality. It processes data using computing units, called *neurons,* arranged into ordered sections, called *layers.* The technique at the foundation of deep learning is the *neural network.*
- **Flexible architectures:** Machine learning solutions offer many knobs (adjustments) called *hyperparameters* that you tune to optimize algorithm learning from data. Deep learning solutions use hyperparameters, too, but they also use multiple user-configured layers (with the user specifying number and type). In fact, depending on the resulting neural network, the number of layers can be quite large and form unique neural networks capable of specialized learning: Some can learn to recognize images, while others can detect and parse voice commands. The point is that the term *deep* is appropriate; it refers to the large number of layers potentially used for analysis. The architecture consists of the ensemble of different neurons and their arrangement in layers in a deep learning solution.
- **Autonomous feature definition:** Machine learning solutions require human intervention to succeed. To process data correctly, analysts and scientists use a lot of their own knowledge to develop working algorithms. For instance, in a machine learning solution that determines the value of a house by relying on data containing the wall measures of different rooms, the machine learning algorithm won't be able to calculate the surface of the house unless the analyst specifies how to calculate it beforehand. Creating the right information for a machine learning algorithm is called feature creation, which is a time-consuming activity. Deep learning doesn't require humans to perform any feature-creation activity because, thanks to its many layers, it defines its own best features. That's also why deep learning outperforms machine learning in otherwise very difficult tasks such as recognizing voice and images, understanding text, or beating a human champion at the Go game (the digital form of the board game in which you capture your opponent's territory).

TECHNICAL STUFF You need to understand a number of issues with regard to deep learning solutions, the most important of which is that the computer still doesn't understand anything and isn't aware of the solution it has provided. It simply provides a form of feedback loop and automation conjoined to produce desirable outputs in less time than a human could manually produce precisely the same result by manipulating a machine learning solution.

The second issue is that some benighted people have insisted that the deep learning layers are hidden and not accessible to analysis. This isn't the case. Anything a computer can build is ultimately traceable by a human. In fact, the General Data Protection Regulation (GDPR) (`https://eugdpr.org/`) requires that humans perform such an analysis (see the article at `https://www.pcmag.com/commentary/361258/how-gdpr-will-impact-the-ai-industry` for details). The requirement to perform this analysis is controversial, but current law says that someone must do it.

The third issue is that self-adjustment goes only so far. Deep learning doesn't always ensure a reliable or correct result. In fact, deep learning solutions can go horribly wrong (see the article at `https://www.theverge.com/2016/3/24/11297050/tay-microsoft-chatbot-racist` for details). Even when the application code doesn't go wrong, the devices used to support the deep learning can (see the article at `https://www.pcmag.com/commentary/361918/learning-from-alexas-mistakes?source=SectionArticles` for details). Despite these problems, you can see deep learning being used for a number of extremely popular applications, as described at `https://medium.com/@vratulmittal/top-15-deep-learning-applications-that-will-rule-the-world-in-2018-and-beyond-7c6130c43b01`.

Performing deep learning tasks

Humans and computers are best at different tasks. Humans are best at reasoning, thinking through ethical solutions, and being emotional. A computer is meant to process data — lots of data — really fast. You commonly use deep learning to solve problems that require looking for patterns in huge amounts of data — problems whose solution is nonintuitive and not immediately noticeable. The article at `http://www.yaronhadad.com/deep-learning-most-amazing-applications/` tells you about 30 different ways in which people are currently using deep learning to perform tasks. In just about every case, you can sum up the problem and its solution as processing huge amounts of data quickly, looking for patterns, and then relying on those patterns to discover something new or to create a particular kind of output.

Employing deep learning in applications

Deep learning can be a stand-alone solution, as illustrated in this book, but it's often used as part of a much larger solution and mixed with other technologies. For example, mixing deep learning with expert systems is not uncommon. The article at `https://www.sciencedirect.com/science/article/pii/0167923694900213` describes this mixture to some degree. However, real applications are more than just numbers generated from some nebulous source. When working in the real world, you must also consider various kinds of data sources and understand how those data sources work. A camera may require a different sort of deep learning solution to obtain information from it, while a thermometer or proximity detector may output simple numbers (or analog data that requires some sort of processing to use). Real-world solutions are messy, so you need to be prepared with more than one solution to problems in your toolkit.

Knowing when not to use deep learning

Deep learning is only one way to perform analysis, and it's not always the best way. For example, even though expert systems are considered old technology, you can’t really create a self-driving car without one for the reasons described at `https://aitrends.com/ai-insider/expert-systems-ai-self-driving-cars-crucial-innovative-techniques/`. A deep learning solution turns out to be way too slow for this particular need. Your car will likely contain a deep learning solution, but you're more likely to use it as part of the voice interface.

REMEMBER AI in general and deep learning in particular can make the headlines when the technology fails to live up to expectations. For example, the article at `https://www.techrepublic.com/article/top-10-ai-failures-of-2016/` provides a list of AI failures, some of which relied on deep learning as well. It’s a mistake to think that deep learning can somehow make ethical decisions or that it will choose the right course of action based on feelings (which no machine has). Anthropomorphizing the use of deep learning will always be a mistake. Some tasks simply require a human.

Speed and the capability to think like a human are the top issues for deep learning, but there are many more. For example, you can’t use deep learning if you don’t have sufficient data to train it. In fact, the article at `https://www.sas.com/en_us/insights/articles/big-data/5-machine-learning-mistakes.html` offers a list of five common mistakes that people make when getting into machine learning and deep learning environments. If you don’t have the right resources, deep learning will never work.

Creating a Pipeline from Data to AI

The previous two sections reveal the need for high-quality and reliable data for any analysis need, especially when it comes to AI, machine learning, and deep learning. In some cases, you need huge quantities of data to make these technologies work correctly. However, before you can do anything, you need a pipeline from the clean source of data you create to the technology used to analyze it. The following sections help you understand these requirements.

Considering the desired output

The point of performing any sort of analysis is to obtain a result that reflects certain characteristics. For example, you may want a probability in one case and a categorization in another. The result might have to be numeric in one case, a string in another, a category in another, or a voice output in another. The output determines all other characteristics of the analysis.

Book 2 tells you about data and how to condition it in various ways. In Book 3, you discover basic analysis techniques that are more machine learning–oriented in nature (or are even basic statistics). Book 4 engages you in simple deep learning. However, the real-world look at output comes in Book 5, in which you see data used in interesting ways, such as working with images and analyzing music and video. In other words, most of this book looks at output first and the input and algorithms to obtain it second.

Defining a data architecture

The emphasis of an analysis model is the data used to feed it. In Book 4, Chapter 1, you begin combining algorithms to obtain various effects. This sort of analysis requires that you supply data in a particular form or else the analysis model you create won't work. So, what you really need is a definition of the data architecture — which is the form in which the data must appear in order to work with the analysis model — before you can massage the raw data. The data architecture reflects the requirements of the pipe through which the data flows to an output.

Combining various data sources

When working in a real-world environment, you seldom find the data you need in one place. The main reason for this issue is that databases store information in a form that reflects the needs of its users — those people who rely on the database for informational needs rather than analysis needs. Another reason is that a data scientist must often look to the results of surveys, generated data, and other analyses for the particulars needed to provide probabilities or perform categorization. Book 2, Chapter 1 helps you understand the need to combine data from various sources into a form that you can use to perform a required analysis.

REMEMBER The use of multiple data sources does mean that you spend a great deal more time ensuring that the data isn't only in the required form, but also that it meshes correctly. You may find that concatenating the datasets won't work — you might have to perform a join, merge, or a cross-tabulation instead. Unless you can visualize how the data should mesh to provide a desired output, the analysis will be flawed, even if you do choose the correct algorithms.

Checking for errors and fixing them

Data errors may not actually appear as errors when you first look at them. In fact, in another situation, the data might not have an error at all. Whether the data is erroneous or not depends on how you use the data and what outcome you need from it. It's not the same as locating an error in code or finding a misplaced symbol in an equation — data errors are grayer, meaning less concrete, than other errors you experience. The grayness of data errors is why Book 6 spends so much time looking at them in multiple ways.

WARNING Never assume that the data does contain an error unless you prove there is an error. The data may simply not perform as you want it to because of errors in format or conditioning. Book 6, Chapter 1 gives you some clues geared toward what you might call *absolute errors* — those that you can possibly prove in some way. Of course, if you decide that the original data is erroneous, fix the perceived error in the original data file, and then later discover that the use or analysis technique is flawed instead, you have damaged your data without any sort of gain. Always work on a copy of your data and make fixes only when you have proven to yourself that the data is truly flawed.

Performing the analysis

Your data pipeline has to remain functional during the analysis if you expect to get the right result. At a minimum, this requirement means that the data remains accessible. Most of the analyses performed in this book rely on static datasets because they're easier to use for example code. However, when you perform real-time analysis, your data pipeline must continue to stream data to the algorithm. In addition, the data pipeline must deliver the data in a timely manner so that the analysis reflects the real-time nature of the application.

Validating the result

To ensure that you get the correct result, you must actually validate the result in some way, which means creating measurable criteria that defines success. Of course, most data scientists know that they have to meet this requirement during the testing phase. A problem can arise, however, when the data, conditions, requirements, or even algorithmic specifications change over time. The changes can occur slowly, but generally you find that the results are marginal on one day and incorrect on the next. Consequently, the need to validate the result isn't a one-time process, but rather is something you continue to do during the life of the application.

Enhancing application performance

When you say that an analysis performs well, the term may mean something different to the listener than it does to you. As with many areas of communication, the meaning of *performance* depends on the biases of the listener and changes in society as a whole (among other things). When looking at analysis performance, you need to consider these issues:

- **Speed:** The analysis is useful only when delivered in a timely manner.
- **Accuracy:** How fast you deliver incorrect information doesn't matter — it's still incorrect.
- **Reliability:** The data pipeline, analysis model, and output mechanisms should perform correctly every time.
- **Predictability:** The result should appear within a defined tolerance unless you find a good reason for a change (such as a difference in conditions).